

those names that could potentially be too expensive to buy and/or sell. One common technique is to limit the position size based on a percentage of the stock's average daily volume, for example, set a maximum size of 10% ADV. The belief in this case is that the market would always be able to absorb an order of this size without the investor inflicting too much impact into the stock price. But the maximum % ADV value is often an arbitrary value and even at this level there are names that are still potentially very expensive to transact.

To better show this point, we analyzed the trading cost corresponding to 10% ADV position size for large (SP500) and small (R2000) cap portfolios. The trading cost for a 10% ADV order size in each of the stocks in the SP500 index is shown in Figure 11.6a. The average cost is 20 bp but these costs vary greatly from a low of 3 bp to a high of 48 bp. For small cap stocks, the average cost of a 10% ADV order is 37 bp with a range of 6 to 160 bp (Figure 11.7a). There are also a very large number of names with costs greater than 60 bp.

Two questions arise when performing this type of PM screening process. First, why should the portfolio managers limit the order size to only 10% ADV for those stocks with very low trading costs? And second, should the maximum size not be set lower for those very expensive to trade stocks? The answer to both of these questions is: Yes. If a stock has low trading costs and is a very appealing investment opportunity the manager should not limit the holding size to some arbitrary value. And if a stock is very expensive to trade, that stock should be held in a much lower quantity in the portfolio—or possibly not held at all.

To help determine the maximum % ADV size to hold in a portfolio, managers can reverse engineer the filtering process. Rather than starting with an arbitrary size constraint, managers can specify the cost level that they feel is appropriate for the portfolio. For example, managers may deem a reasonable cost to trade large cap stocks is 20 bp. This cost level is often tied to the expected alpha of the fund and will be further discussed below. Then managers compute for the investable universe the size that can be traded resulting in the cost of 20 bp. The maximum size for the stock is then determined for all stocks by solving the MI equation by setting the LHS equal to 20 bp and solving for size. Mathematically, this is found by solving for size in the following:

$$20 = b_1 \cdot (a_1 \cdot \text{Size}^{a_2} \cdot \sigma^{a_3}) \cdot \text{POV}^{a_4} + (1 - b_1) \cdot (a_1 \cdot \text{Size}^{a_2} \cdot \sigma^{a_3}) \quad (11.9)$$